# Relations & Equivalence

## 2.AB PreIB Maths – Exam D

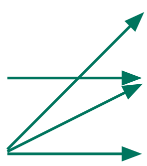
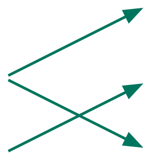
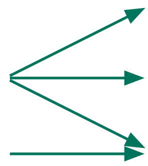
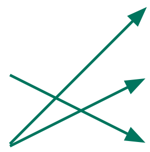
> Unless specified otherwise, you are to **always** (at least briefly) explain your reasoning. Even in closed questions.

### Product of Sets & Relations

Let $A = \{\ ,\ \}$ and $B = \{\ ,\ ,\ \}$. Check all **correct** statements of the ones below. You **don't have to** explain yourself. [25 %]

- ☐ The set $\{(\ ,\ ), (\ ,\ ), (\ ,\ )\}$ is a relation from $B$ to $A$.
- ☐ The set $(A \times B) \cap (B \times A)$ is **empty**.
- ☐ There are **more** relations from $B$ to $A$ than there are relations from $A$ to $B$.
- ☐ The element $(\ ,\ )$ lies in $A \times A$.
- ☐ The set $B \times B$ has 9 elements.

A relation $R$ from $A$ to $B$ is called **injective** if whenever $a_1 R b$ and $a_2 R b$, then $a_1 = a_2$ for $a_1, a_2 \in A$ and $b \in B$. In words, no two **different** elements from $A$ can be related to the **same** element in $B$. Mark only those relations below that are injective. [25 %]

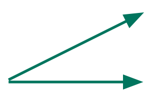
### Bonus Problem

The **union** of relations $R$ and $S$ from $A$ to $B$ is the relation written as $R \cup S$ from $A$ to $B$ that is defined by [10 %]

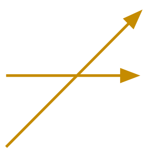
$$a(R \cup S)b \text{ in the case that } aRb \text{ or } aSb.$$

for any elements $a \in A, b \in B$. Draw the picture for the relation $R \cup S$ if the pictures of $R$ and $S$ are as below.

$R$



$S$



Draw $R \cup S$ here.

## Equivalence

Recall that the relation of *equivalence* is given by three conditions: [25 %]

- **reflexivity**: every element is equivalent to itself;
- **symmetry**: if $a$ is equivalent to $b$, then $b$ is equivalent to $a$;
- **transitivity**: if $a$ is eq. to $b$ and $b$ is eq. to $c$, then $a$ is eq. to $c$.

Determine if the relation

$$R = \{(1,1), (1,2), (2,2), (3,2), (4,4), (4,1)\}$$

is an equivalence on the set $A = \{1,2,3,4,5\}$. If not, encircle all the pairs below **that must be added** to $R$ because of one of the rules above.

| | | | | |
|---|---|---|---|---|
| $(3,3)$ | $(2,3)$ | $(1,5)$ | $(5,2)$ | $(5,5)$ |
| $(2,4)$ | $(1,3)$ | $(3,4)$ | $(1,4)$ | $(5,1)$ |

Describe using a picture an equivalence on the set $A = \{1,2,3,4,5\}$ which has the following **classes of equivalence**: [25 %]

- $\{1,2\}$, $\{3,4\}$ and $\{5\}$.
- $\{1,5\}$ and $\{2,3,4\}$.

## Bonus Problem

Draw a relation on the set $A = \{1,2,3,4,5\}$, which is [10 %]

(a) symmetric and transitive but **not** reflexive.

(b) reflexive and transitive but **not** symmetric.

| (a) | | (b) | |
|---|---|---|---|
| 5 | 5 | 5 | 5 |
| 4 | 4 | 4 | 4 |
| 3 | 3 | 3 | 3 |
| 2 | 2 | 2 | 2 |
| 1 | 1 | 1 | 1 |